UNIVERSITI TUNKU ABDUL RAHMAN
Faculty of Information and Communication Technology

UCCD3223 Mobile Applications Development
(June 2024 Trimester)

Individual Practical Assignment

Name	**TAN ZONG TING**
Student ID	**23ACB02731**
Course	**CN**
Practical Group	**P2**
Lecturer	**Mr. Tan Chiang Kang @ Thang Chiang Kang**

Marking scheme	Marks	Remarks
Correctness of output	× 2.5	
Design of UI	× 2.5	
User Friendliness	× 3	
Neat Program Documentation		
Report Format		
TOTAL		

Homepage of Loan Calculate

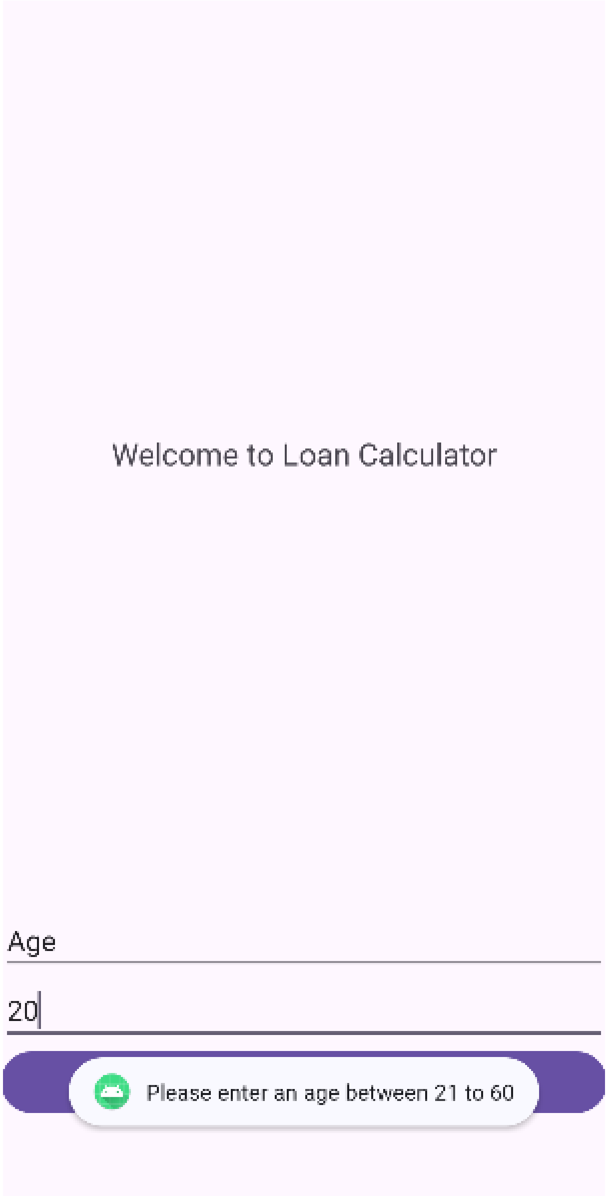

Figure 1

According to the bank in Malaysia, the minimum age for having personal loan or any loan is 21 years old and above and maximum of the age will be 60 years old. Therefore, when the user enter age below 21 and above 60. It will pop out an alert message (***Figure 1***). After popped out the message, user can enter the correct age again. Then it will pop out an alert message "Submitted successfully" shown as ***Figure x***.

Selection Page of Loan Calculate

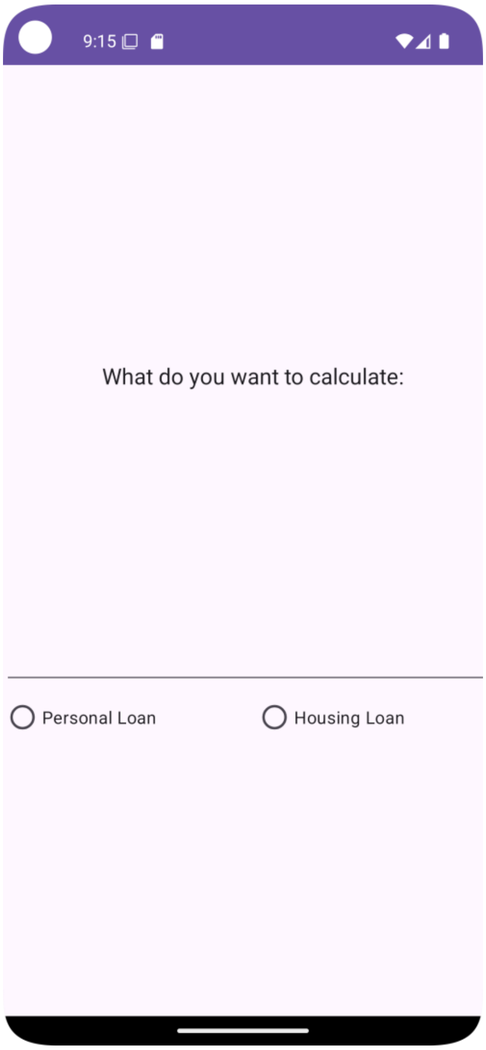

Figure 2

After user enter their age on the Figure 1, it will proceed to the next page will is the selection as shown as ***Figure 2***. In this section, user may choose one of the following which is Personal Loan and Housing Loan.

Personal Loan Calculation page

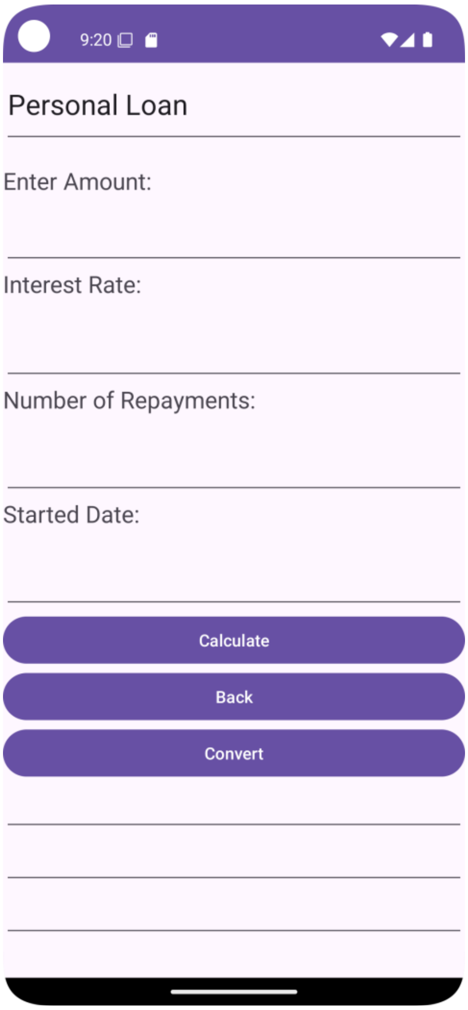

Figure 3

When the user clicked on the Personal Loan button in ***Figure 2***, it will proceed to personal loan calculation page. Then the user can enter the loan amount, interest rate, number of repayments and the started date which provides in ***Figure 3***.

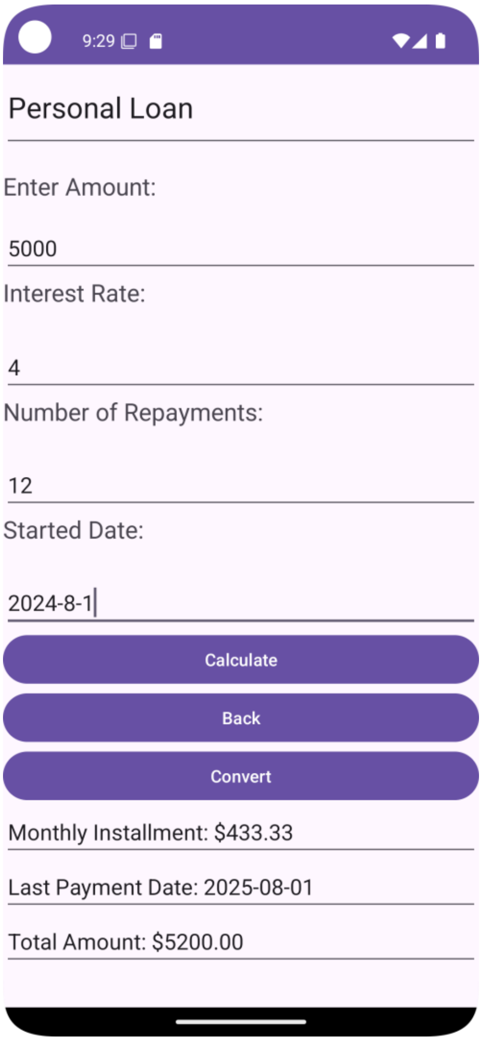

Figure 4

For example, if the user enters total amount RM5000, interest rate 4%, number of repayments 12 and the started date 1st August 2024. After clicking the "Calculate" button, the result of monthly instalment, last payment date and total amount will be shown at the bottom of the third button. (***Figure 4***)

Table of Amortization Schedule table (personal loan)

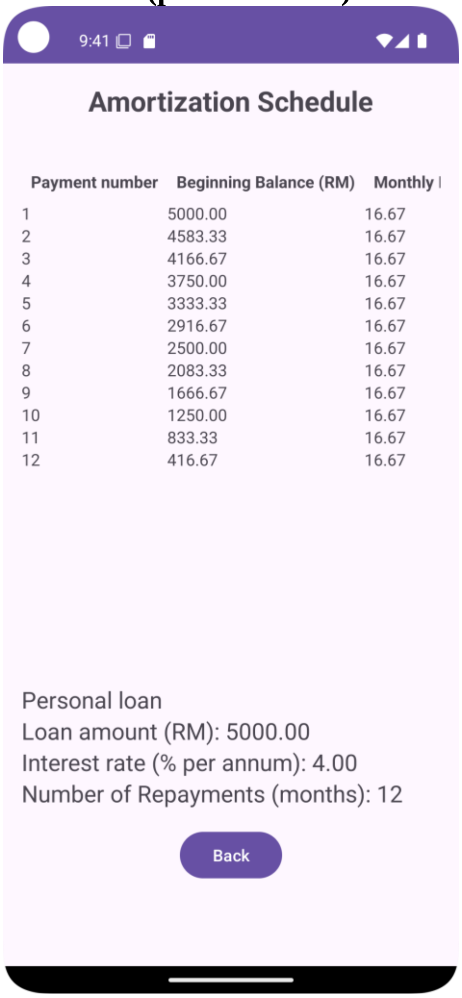

Figure 5

When the user clicked on "Convert" button, it will lead them to the amortization schedule table shown above. This amortization schedule will show the payment number, beginning balance (RM), Monthly payment (RM), Interest paid (RM), and Principal paid (RM). This table have a function which is scrolling up and down, left and right. It has a "Back" button which can lead the user back to the personal loan in ***Figure 4***.

Housing Loan Calculation page

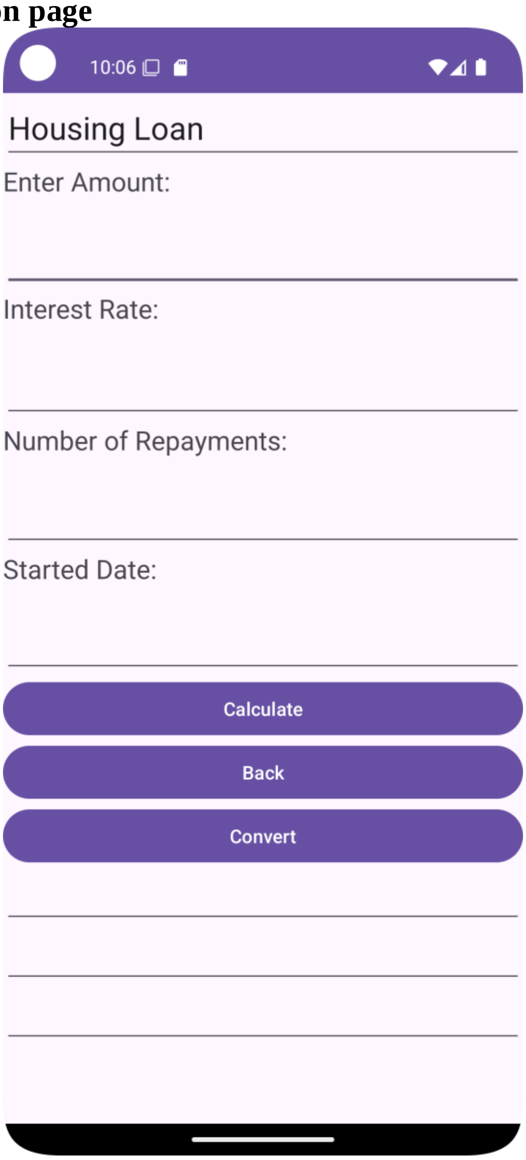

Figure 6

When the user clicked on the Housing Loan button in ***Figure 2***, it will proceed to housing loan calculation page. Then the user can enter the loan amount, interest rate, number of repayments and the started date which provides in ***Figure 6***.

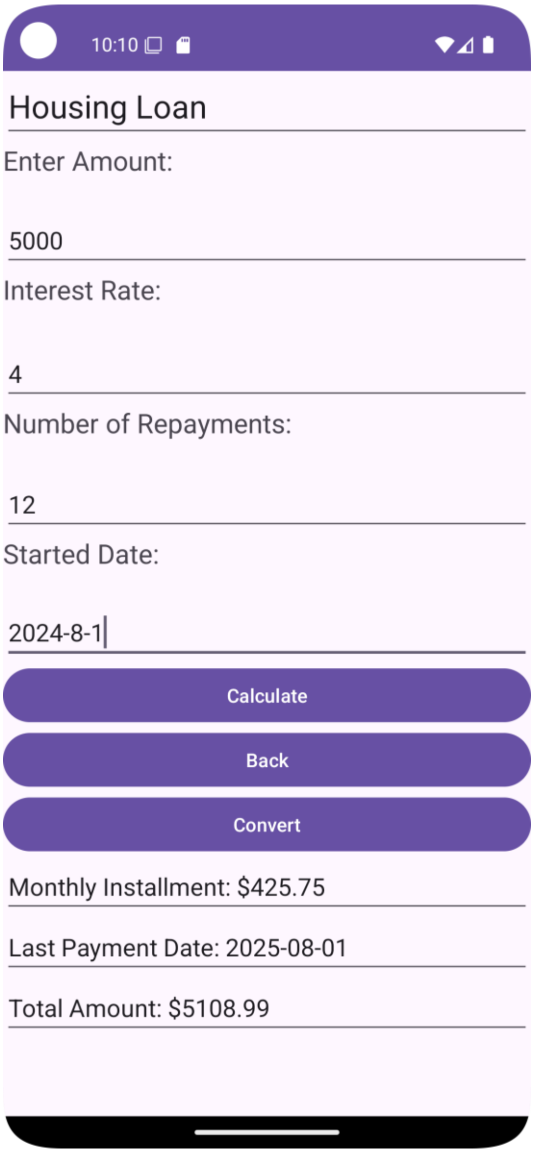

Figure 7

For example, if the user enters total amount RM5000, interest rate 4%, number of repayments 12 and the started date 1st August 2024. After clicking the "Calculate" button, the result of monthly instalment, last payment date and total amount will be shown at the bottom of the third button. (***Figure 7***)

Table of Amortization Schedule table (housing loan)

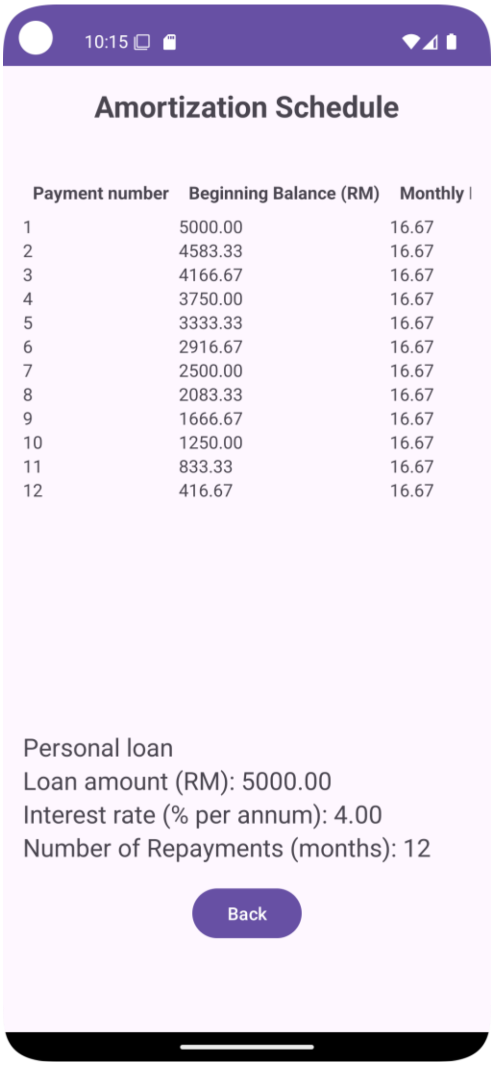

Figure 8

When the user clicked on "Convert" button, it will lead them to the amortization schedule table shown above. This amortization schedule will show the payment number, beginning balance (RM), Monthly payment (RM), Interest paid (RM), and Principal paid (RM). This table have a function which is scrolling up and down, left and right. It has a "Back" button which can lead the user back to the personal loan in ***Figure 8.***

Appendix

Welcome to Loan Calculate.kt

```kotlin
package com.example.loancalculator

import android.content.Intent
import androidx.appcompat.app.AppCompatActivity
import android.os.Bundle
import android.widget.Button
import android.widget.EditText
import android.widget.Toast
import com.example.loancalculator.R.id.bt_nextpg
import com.example.loancalculator.R.id.tx_date

class MainActivity : AppCompatActivity() {
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)

        val dateBtn = findViewById<Button>(bt_nextpg)
        dateBtn.setOnClickListener {
            val dateText = findViewById<EditText>(tx_date).text.toString()
            val age = dateText.toIntOrNull()

            if (dateText.isEmpty()) {
                Toast.makeText(this, "Please fill in your age!", Toast.LENGTH_SHORT).show()
            } else if (age == null || age < 22 || age > 60) {
                Toast.makeText(this, "Please enter an age between 22 to 60", Toast.LENGTH_SHORT).show()
            } else {
                Toast.makeText(this, "Submitted successfully!", Toast.LENGTH_SHORT).show()
                val intent1 = Intent(this, MainActivity2::class.java)
                startActivity(intent1)
            }
        }
    }
}
```

Selection.kt

```kotlin
package com.example.loancalculator

import android.content.Intent
import androidx.appcompat.app.AppCompatActivity
import android.os.Bundle
import android.widget.Button

class MainActivity2 : AppCompatActivity() {
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main2)


        var button2 = findViewById<Button>(R.id.tx_personalLoan)
        button2.setOnClickListener(){
            val Intent2 = Intent(this, MainActivity3::class.java)
            startActivity(Intent2)
        }

        var button3 = findViewById<Button>(R.id.tx_housingLoan)
        button3.setOnClickListener(){
            val Intent3 = Intent(this, MainActivity4::class.java)
            startActivity(Intent3)
        }
    }
}
```

Personal Loan.kt

```kotlin
package com.example.loancalculator
import android.content.Intent
import androidx.appcompat.app.AppCompatActivity
import android.os.Bundle
import android.widget.Button
import android.widget.EditText
import android.widget.TextView
import android.widget.Toast
import java.text.SimpleDateFormat
import java.util.Calendar
import java.util.Locale

class MainActivity3 : AppCompatActivity() {
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main3)

        val button4 = findViewById<Button>(R.id.bt_back)
        button4.setOnClickListener {
            Toast.makeText(this, "Back button clicked", Toast.LENGTH_SHORT).show()
            val intent3 = Intent(this, MainActivity2::class.java)
            startActivity(intent3)
        }

        val amountEditText: EditText = findViewById(R.id.line2)
        val interestRateEditText: EditText = findViewById(R.id.line4)
        val numInstallmentsEditText: EditText = findViewById(R.id.line6)
        val startDateEditText: EditText = findViewById(R.id.line8)
        val calculateButton: Button = findViewById(R.id.bt_calculate)
        val monthlyInstallmentTextView: TextView = findViewById(R.id.tx_outputanswer)
        val lastPaymentDateTextView: TextView = findViewById(R.id.line10)
        val totalAmountTextView: TextView = findViewById(R.id.line12)

        calculateButton.setOnClickListener {
            Toast.makeText(this, "Calculate button clicked", Toast.LENGTH_SHORT).show()
            val amount = amountEditText.text.toString().toDoubleOrNull()
            val interestRate = interestRateEditText.text.toString().toDoubleOrNull()
            val numInstallments = numInstallmentsEditText.text.toString().toIntOrNull()
            val startDate = startDateEditText.text.toString()

            if (amount != null && interestRate != null && numInstallments != null && startDate.isNotEmpty()) {
                val monthlyInstallment = calculateMonthlyInstallment(amount, interestRate, numInstallments)
                val lastPaymentDate = calculateLastPaymentDate(startDate, numInstallments)
                val totalAmount = calculateTotalAmount(amount, interestRate, numInstallments)

                monthlyInstallmentTextView.text = "Monthly Installment: $%.2f".format(monthlyInstallment)
                lastPaymentDateTextView.text = "Last Payment Date: $lastPaymentDate"
                totalAmountTextView.text = "Total Amount: $%.2f".format(totalAmount)

                // Pass data to MainActivity5
                val intent = Intent(this, MainActivity5::class.java)
                intent.putExtra("loanAmount", amount)
                intent.putExtra("interestRate", interestRate)
                intent.putExtra("months", numInstallments)
                startActivity(intent)
            } else {
                Toast.makeText(this, "Please fill in all fields correctly", Toast.LENGTH_SHORT).show()
            }
        }
    }

    private fun calculateMonthlyInstallment(amount: Double, interestRate: Double, numInstallments: Int): Double {
        val monthlyInterestRate = (interestRate / 100) / 12
        return (amount * (1 + (monthlyInterestRate * numInstallments))) / numInstallments
    }

    private fun calculateLastPaymentDate(startDate: String, numInstallments: Int): String? {
        val sdf = SimpleDateFormat("yyyy-MM-dd", Locale.getDefault())
        val date = sdf.parse(startDate)
        return if (date != null) {
            val calendar = Calendar.getInstance()
            calendar.time = date
            calendar.add(Calendar.MONTH, numInstallments)
            sdf.format(calendar.time)
        } else {
            null
        }
    }

    private fun calculateTotalAmount(amount: Double, interestRate: Double, numInstallments: Int): Double {
        val monthlyInstallment = calculateMonthlyInstallment(amount, interestRate, numInstallments)
        return monthlyInstallment * numInstallments
    }
}
```

Housing Loan.kt

```kotlin
package com.example.loancalculator

import android.content.Intent
import androidx.appcompat.app.AppCompatActivity
import android.os.Bundle
import android.widget.Button
import android.widget.EditText
import android.widget.TextView
import android.widget.Toast
import java.text.SimpleDateFormat
import java.util.Calendar
import java.util.Locale
import kotlin.math.pow

class MainActivity4 : AppCompatActivity() {
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main4)

        val button5 = findViewById<Button>(R.id.bt_back2)
        button5.setOnClickListener {
            val intent4 = Intent(this, MainActivity2::class.java)
            startActivity(intent4)
        }

        val button7 = findViewById<Button>(R.id.bt_gnrate2)
        button7.setOnClickListener {
            Toast.makeText(this, "Convert button clicked", Toast.LENGTH_SHORT).show()
            val intent6 = Intent(this, MainActivity6::class.java)
            startActivity(intent6)
        }

        val amountEditText: EditText = findViewById(R.id.line22)
        val interestRateEditText: EditText = findViewById(R.id.line44)
        val numInstallmentsEditText: EditText = findViewById(R.id.line66)
        val startDateEditText: EditText = findViewById(R.id.line88)
        val calculateButton: Button = findViewById(R.id.Calculatebutton)
        val monthlyInstallmentTextView: TextView = findViewById(R.id.tx_outputanswer2)
        val lastPaymentDateTextView: TextView = findViewById(R.id.line100)
        val totalAmountTextView: TextView = findViewById(R.id.line122)

        calculateButton.setOnClickListener {
            Toast.makeText(this, "Calculate button clicked", Toast.LENGTH_SHORT).show()
            val amount = amountEditText.text.toString().toDoubleOrNull()
```

```kotlin
            val interestRate = interestRateEditText.text.toString().toDoubleOrNull()
            val numInstallments = numInstallmentsEditText.text.toString().toIntOrNull()
            val startDate = startDateEditText.text.toString()

            if (amount != null && interestRate != null && numInstallments != null && startDate.isNotEmpty()) {
                val monthlyInstallment = calculateMonthlyInstallment(amount, interestRate, numInstallments)
                val lastPaymentDate = calculateLastPaymentDate(startDate, numInstallments)
                val totalAmount = calculateTotalAmount(amount, interestRate, numInstallments) // Calculate total amount

                monthlyInstallmentTextView.text = "Monthly Installment: $%.2f".format(monthlyInstallment)
                lastPaymentDateTextView.text = "Last Payment Date: $lastPaymentDate"
                totalAmountTextView.text = "Total Amount: $%.2f".format(totalAmount)

                val intent = Intent(this, MainActivity6::class.java)
                intent.putExtra("loanAmount", amount)
                intent.putExtra("interestRate", interestRate)
                intent.putExtra("months", numInstallments)
                startActivity(intent)

            } else {
                Toast.makeText(this, "Please fill in all fields correctly", Toast.LENGTH_SHORT).show()
            }
        }
    }

    private fun calculateMonthlyInstallment(
        amount: Double,
        interestRate: Double,
        numInstallments: Int
    ): Double {
        val monthlyInterestRate = (interestRate / 100) / 12
        return if (monthlyInterestRate == 0.0) {
            amount / numInstallments // Handle zero interest rate
        } else {
            val factor = (1 + monthlyInterestRate).pow(numInstallments)
            (amount * monthlyInterestRate * factor) / (factor - 1)
        }
    }

    private fun calculateLastPaymentDate(startDate: String, numInstallments: Int): String? {
        val sdf = SimpleDateFormat("yyyy-MM-dd", Locale.getDefault())
        val date = sdf.parse(startDate)
```

```kotlin
        return if (date != null) {
            val calendar = Calendar.getInstance()
            calendar.time = date
            calendar.add(Calendar.MONTH, numInstallments)
            sdf.format(calendar.time)
        } else {
            null
        }
    }

    private fun calculateTotalAmount(amount: Double, interestRate: Double, numInstallments: Int): Double {
        val monthlyInstallment = calculateMonthlyInstallment(amount, interestRate, numInstallments)
        return monthlyInstallment * numInstallments
    }
}
```

Amortization Schedule (personal).kt

```kotlin
package com.example.loancalculator

import androidx.appcompat.app.AppCompatActivity
import android.os.Bundle
import android.widget.Button
import android.widget.TableLayout
import android.widget.TableRow
import android.widget.TextView

class MainActivity5 : AppCompatActivity() {
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main5)

        val tableLayout: TableLayout = findViewById(R.id.amortizationTable)
        val backBtn: Button = findViewById(R.id.backToCalculatorBtn)

        val loanAmountGenerate: TextView = findViewById(R.id.loanAmountGenerate)
        val interestGenerate: TextView = findViewById(R.id.interestGenerate)
        val loanTenureGenerate: TextView = findViewById(R.id.loanTenureGenerate)

        // Receive data from the intent
        val principal = intent.getDoubleExtra("loanAmount", 0.0)
        val interestRate = intent.getDoubleExtra("interestRate", 0.0)
        val months = intent.getIntExtra("months", 0)

        // Display data
        loanAmountGenerate.text = String.format("Loan amount (RM): %.2f", principal)
        interestGenerate.text = String.format("Interest rate (%% per annum): %.2f", interestRate)
        loanTenureGenerate.text = String.format("Number of Repayments (months): %d", months)

        val amortizationSchedule = generateAmortization(principal, interestRate, months)

        for (month in amortizationSchedule) {
            val row = TableRow(this)
            val monthView = TextView(this)
            val beginningBalanceView = TextView(this)
            val interestPaidView = TextView(this)
            val principalPaidView = TextView(this)
            val monthlyPaymentView = TextView(this)

            monthView.text = month.month.toString()
            beginningBalanceView.text = String.format("%.2f", month.beginningBalance)
            interestPaidView.text = String.format("%.2f", month.interestPaid)
```

```
            principalPaidView.text = String.format("%.2f", month.principalPaid)
            monthlyPaymentView.text = String.format("%.2f", month.monthlyPayment)

            row.addView(monthView)
            row.addView(beginningBalanceView)
            row.addView(interestPaidView)
            row.addView(principalPaidView)
            row.addView(monthlyPaymentView)
            tableLayout.addView(row)
        }

        backBtn.setOnClickListener {
            finish()
        }
    }

    private fun generateAmortization(principal: Double, interestRate: Double, months: Int): List<AmortizationMonth> {
        val amortizationSchedule = mutableListOf<AmortizationMonth>()
        val monthlyRate = (interestRate / 100) / 12
        val monthlyPayment = (principal + (principal * (interestRate / 100))) / months
        val interestPaid = principal * monthlyRate
        val principalPaid = monthlyPayment - interestPaid

        var remainingPrincipal = principal

        for (i in 1 .. months) {
            val beginningBalance = remainingPrincipal
            remainingPrincipal -= principalPaid

            amortizationSchedule.add(AmortizationMonth(i, beginningBalance, interestPaid, principalPaid, monthlyPayment))
        }

        return amortizationSchedule
    }

    data class AmortizationMonth(
        val month: Int,
        val beginningBalance: Double,
        val interestPaid: Double,
        val principalPaid: Double,
        val monthlyPayment: Double
    )
```

Amortization Schedule (housing).kt

```kotlin
package com.example.loancalculator

import androidx.appcompat.app.AppCompatActivity
import android.os.Bundle
import android.widget.Button
import android.widget.TableLayout
import android.widget.TableRow
import android.widget.TextView

class MainActivity6 : AppCompatActivity() {
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main6)


        val tableLayout: TableLayout = findViewById(R.id.amortizationTable)
        val backBtn: Button = findViewById(R.id.backToCalculatorBtn2)

        val loanAmountGenerate: TextView = findViewById(R.id.loanAmountGenerate2)
        val interestGenerate: TextView = findViewById(R.id.interestGenerate2)
        val loanTenureGenerate: TextView = findViewById(R.id.loanTenureGenerate2)

        // Receive data from the intent
        val principal = intent.getDoubleExtra("loanAmount", 0.0)
        val interestRate = intent.getDoubleExtra("interestRate", 0.0)
        val months = intent.getIntExtra("months", 0)

        // Display data
        loanAmountGenerate.text = String.format("Loan amount (RM): %.2f", principal)
        interestGenerate.text = String.format("Interest rate (%% per annum): %.2f", interestRate)
        loanTenureGenerate.text = String.format("Number of Repayments (months): %d", months)

        val amortizationSchedule = generateAmortization(principal, interestRate, months)

        for (month in amortizationSchedule) {
            val row = TableRow(this)
            val monthView = TextView(this)
            val beginningBalanceView = TextView(this)
            val interestPaidView = TextView(this)
            val principalPaidView = TextView(this)
            val monthlyPaymentView = TextView(this)

            monthView.text = month.month.toString()
            beginningBalanceView.text = String.format("%.2f", month.beginningBalance)
            interestPaidView.text = String.format("%.2f", month.interestPaid)
            principalPaidView.text = String.format("%.2f", month.principalPaid)
            monthlyPaymentView.text = String.format("%.2f", month.monthlyPayment)

            row.addView(monthView)
            row.addView(beginningBalanceView)
            row.addView(interestPaidView)
            row.addView(principalPaidView)
            row.addView(monthlyPaymentView)
            tableLayout.addView(row)
        }

        backBtn.setOnClickListener {
            finish()
        }
    }

    private fun generateAmortization(principal: Double, interestRate: Double, months: Int): List<AmortizationMonth> {
        val amortizationSchedule = mutableListOf<AmortizationMonth>()
        val monthlyRate = (interestRate / 100) / 12
        val monthlyPayment = (principal + (principal * (interestRate / 100))) / months
        val interestPaid = principal * monthlyRate
        val principalPaid = monthlyPayment - interestPaid

        var remainingPrincipal = principal

        for (i in 1..months) {
            val beginningBalance = remainingPrincipal
            remainingPrincipal -= principalPaid

            amortizationSchedule.add(AmortizationMonth(i, beginningBalance, interestPaid, principalPaid, monthlyPayment))
        }

        return amortizationSchedule
    }

    data class AmortizationMonth(
        val month: Int,
        val beginningBalance: Double,
        val interestPaid: Double,
        val principalPaid: Double,
        val monthlyPayment: Double
    )
}
```